

CT-EBM-SP - Corpus of Clinical Trials for Evidence-Based-Medicine in Spanish (version 3)

Project documentation

Companion file to article:

Transformer-based relation extraction and concept normalization using an annotated clinical trials corpus

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Related Annotation Work

Corpus	Text type and size	Normalizations (count)
BioCreative ¹	6218 journal abstracts.	Normalization to yeast, fly and mouse genes identifiers.
CANTEMIST ²	1301 anonymized oncology clinical cases.	16030 entities normalized to the ICD-O-3 tumour classification (850 unique codes).
DisTEMIST ³	1000 anonymized clinical cases extracted from journal articles.	10 665 entities normalized to SNOMED-CT (7303 unique codes).
NCBI corpus ⁴	793 abstracts.	Disease entities (6892) normalized to MeSH and OMIM.
CRAFT ⁵	67 full-text journal articles.	Nearly 100 000 concepts from the Gene Ontology.
ShARe (SemEval 2015 Task 14 ⁶)	531 clinical notes.	Disorder entities (19 111) normalized to the UMLS.
MCN corpus ⁷	100 discharge summaries.	10 919 concept mentions normalized to the UMLS.
QUAERO ⁸	2500 Medline titles, 13 texts from EMEA, and 25 EPO patents in French	26 409 entity annotations were mapped to 26 281 UMLS CUIs (5797 unique codes).
MANTRA ⁹	1450 texts from EMEA, Medline and biomedical patent claims in English, French, German, Spanish and Dutch.	5530 annotations (total for all languages) normalized to the UMLS.
E3C ¹⁰	422 anonymized clinical cases extracted from journal articles in Basque, English, French, Italian and Spanish.	6475 entities (total for all languages) normalized to the UMLS.

Acronyms: *CT*: clinical trials; *CUI*: Concept Unique Identifier; *E3C*: European Clinical Case Corpus; *EMEA*: European Medicines Agency; *ICD-O-3*: International Classification of Diseases for Oncology vs 3; *EPO*: European Patent Office; *MCN*: Medical Concept Normalization; *MeSH*: Medical Subject Headings; *NCBI*: National Center for Biotechnology Information; *OMIM*: Online Mendelian Inheritance in Man; *SNOMED-CT*: Systematized Nomenclature of Medicine – Clinical Terms; *UMLS*: Unified Medical Language System.

Table 1. Biomedical and CT-related corpora for medical concept normalization.

Corpus	Text type and size	Annotations (count)
i2b2 ¹¹	Patient reports (394 in train + 477 in test).	3 entity types (in total, 27 837 train, 5009 test), and 6 relation types (5264 train; 9069 test), as reported in ¹² .
ADE ¹³	Medical reports (4272 sentences).	3 entity types: Drug (5063), Dosage (231) and Adverse-Effect (5776); 2 relation types: Adverse-Effect (6821) and Drug-dosage (279).
DDI ¹⁴	1025 documents from DrugBank and Medline.	4 entity types (18 502) and 4 types of DDI relationships (5028).
CDR ¹⁵	1500 abstracts from Pubmed.	Chemical entities (4409), Disease entities (5818) and Chemical-Disease relations (3116).
n2c2 2018 ADE ¹⁶	505 discharge summaries.	9 entity types (83 869) and 8 relation types (59 810).
Chia ¹⁷	1000 texts from ClinicalTrials.gov (12 409 eligibility criteria).	15 entity types (41 487) and 12 different relationships (25 017).
Tseo et al. (2020) ¹⁸	3314 trials from ClinicalTrials.gov	10 clinical entity types, 5 attributes and value limits (121 221); negation relation (5054).
Tseo et al. (2020) ¹⁹	470 study protocols from a private database.	15 clinical entity types and 7 relation types.
Nye et al. (2021) ²⁰	1932 journal abstracts describing CTs.	Entities describing interventions and outcomes (13895), and relations between them (5054).
BioRED ²¹	600 PubMed abstracts.	6 entity types (20 419), 8 relationships (6503) and relation pairs with novelty findings (4532).
SNPPhenA ²²	360 abstracts from PubMed.	SNPs (868), phenotypes (590), SNP-Phenotype association candidates (1300), neutral (308), negative (120) and positive candidates (872); negation and modality markers.
DGO ²³	1622 oncology exams and breast cancer notes in Portuguese.	10 entity types (146 769) and 3 relation types (111 716).
LEAF ²⁴	1006 CT descriptions from ClinicalTrials.gov (105 816 annotations).	50 entity types (41 487) and 51 relation types (25 017).
RCT-ART ²⁵	558 sentences from 6 disease areas.	3 entity types (3541) and 3 relationships (3182).

Table 2. Biomedical and CT-related corpora annotated with relationships.

Acronyms: *ADE*: Adverse Drug Effects; *CDR*: Chemical-Disease Relations; *CTs*: Clinical Trials; *DDI*: Drug-Drug Interactions; *SNPs*: Single Nucleotide Polymorphisms.

Annotation Scheme

UMLS	ANAT: Anatomical structure, body substance or cell.	<i>músculo</i> ('muscle')
	CHEM: Chemical or pharmacological substance.	<i>antibiótico</i> ('antibiotic')
	DEVI: Medical device.	<i>sonda</i> ('probe')
	DISO: Pathology, disorder, neoplasms, injuries or signs and symptoms.	<i>cancer, fiebre</i> ('fever')
	GENE: Nucleotide sequence, gene or genome.	<i>HER2, BRAF</i>
	LIVB: Human; professional or population group; animal, bacterium, fungus; plant and virus.	<i>paciente</i> ('patient'), <i>S. aureus</i>
	PHYS: Biologic, cell, genetic or physiologic function.	<i>digestión</i> ('digestion')
	PROC: Diagnostic, therapeutic or laboratory procedure; health care activity; research activity.	<i>hemograma</i> ('hemogram'), <i>bypass</i>
Intervention	Contraindicated: Any procedure or drug that is a contraindication. [Attribute]	<i>Contraindicación a AAS</i> (('ASA contraindication'))
	Dose_or_Strength: Drug dose or strength.	<i>2mg, 2%</i>
	Form: Dosage form.	<i>pastilla</i> ('pill')
	Route_or_Mode: Route or mode of administration.	<i>per os</i> ('oral')
Temporal	Age. [entity or attribute]	<i>18 años</i> ('18 y.o.')
	Date: Specific or relative date.	<i>2020, hoy</i> ('today')
	Duration_or_Interval: length of time that an event lasts.	<i>6 meses</i> ('6 months')
	Frequency: How often an event occurs.	<i>semanal</i> ('weekly')
	Time: Hour or part of the day.	<i>15:00</i>
Assertion	Neg_cue: Negation cue.	<i>no</i>
	Negated: Negated concept or event. [Attribute]	<i>sin fiebre</i> ('no fever')
	Spec_cue: Speculation cue.	<i>posible</i> ('possible')
	Speculated: Speculated concept or event. [Attribute]	<i>probable diabetes</i>
Event temporality	Family_History_of: Record of medical conditions or procedures in a patient's family. [Attribute]	<i>Antecedentes familiares: DB</i> (('Family history: DB'))
	Future: Any expected event or a pending procedure. [Attribute]	<i>embarazo planificado</i> (('planned pregnancy'))
	History_of: Record of medical conditions or procedures. [Attribute]	<i>Antecedentes de ictus</i> (('history of stroke'))
	Hypothetical: Any event that is imagined, but it is not truly happening at the moment. [Attribute]	<i>si es mujer, debe...</i> (('If female, subject must...'))
Experiencer	Family_member: A patient's relative. [Attribute]	<i>padre</i> ('father')
	Other: Any person that is not the patient/participant nor a relative. [Attribute]	<i>representante legal</i> (('legal representative'))
	Patient: ill person or participant in a trial. [Attribute]	<i>pacientes pediátricos</i> (('pediatric patients'))
Other	Concept: idea, evaluation scale, procedure approach or temporal concept not in TimeML.	<i>esperanza de vida</i> (('life expectancy'))
	Food_or_Drink	<i>soja</i> ('soy'), <i>alcohol</i>
	Observation_or_Finding: Clinical findings or conditions that are not pathological states.	<i>normotenso</i> (('normal pressure'))
	Quantifier_or_Qualifier: modifiers/adjectives needed to understand the text.	<i>grave</i> ('severe')
	Result_or_Value: Outcome of a lab or a procedure.	<i>> 140 kg</i>

Table 3. Annotated entities and attributes with examples.

	Relation type	Examples
Assertion	Negation	<i>no</i> → <i>fiebre</i> (‘no fever’)
	Speculation	<i>posible</i> → <i>diabetes</i> (‘possible diabetes’)
Intervention-related / temporal	Has_Dose_or_Strength	<i>aspirina</i> → <i>500 mg</i>
	Has_Form	<i>aspirina</i> → <i>comprimido</i> (‘tablet’)
	Has_Route_or_Mode	<i>aspirina</i> → <i>oral</i>
	Has_Duration_or_Interval	<i>aspirina</i> → <i>un mes</i> (‘one month’)
	Has_Frequency	<i>aspirina</i> → <i>diaria</i> (‘daily’)
	Has_Age	<i>paciente</i> (‘patient’) → <i>50 años</i> (‘50 y.o.’)
	After	<i>sutura</i> (‘suture’) → <i>operación</i> (‘surgery’)
	Before	<i>anestesia</i> (‘anesthesia’) → <i>operación</i> (‘surgery’)
	Overlap	<i>operación</i> (‘surgery’) → <i>2009</i>
	Combined_with	<i>dolutegravir</i> → <i>lamivudina</i> (‘lamivudine’)
	Used_for	<i>aspirina</i> → <i>tratamiento</i> (‘treatment’)
	Causes	<i>VIH</i> (‘HIV’) → <i>SIDA</i> (‘AIDS’)
Other	Experiences	<i>paciente</i> (‘patient’) → <i>diabetes</i>
	Has_Quantifier_or_Qualifier	<i>asma</i> (‘asthma’) → <i>grave</i> (‘severe’)
	Has_Result_or_Value	<i>IMC</i> (‘BMI’) → <i>> 25</i>
	Location_of	<i>dolor</i> (‘pain’) → <i>oído</i> (‘ear’)

Table 4. Annotated relations with examples.

Annotation Results

We annotated 87037 entities, including 14518 nested entities (16.68%) and 557 (0.64%) discontinuous entities; 16597 attributes and 68217 relationships. 81.75% of entities were normalized. Tables 5 and 6 report the count of entities, attributes and relations. Table 7 shows the count of normalized entities, including the unclear normalization (i.e., entities marked with the '?' character) and the complex normalization (i.e., entities mapped to two or more UMLS CUIs). Figure 1 also shows a chord diagram with the 20 most frequent relations between entities (we used the normalized CUIs to count the relations, and showed the English preferred term).

	Abstracts	EudraCT	Total
ANAT	2780	4204	6984
CHEM	4391	4992	9383
DEVI	451	193	644
DISO	4325	8894	13219
GENE	16	75	91
LIVB	3432	4951	8383
PHYS	887	1480	2367
PROC	9276	10120	19396
Contraindicated	4	263	267
Dose_or_Strength	875	223	1098
Form	167	70	237
Route_or_Mode	548	444	992
Age	528	1051	1579
Date	606	1301	1907
Duration_or_Interval	1421	1543	2964
Frequency	300	126	426
Time	343	72	415
Concept	2298	1073	3371
Food_or_Drink	146	44	190
Observation	1724	2174	3898
Quantifier_or_Qualifier	1856	3108	4964
Result_or_Value	216	1409	1625
Neg_cue	1195	1564	2759
Negated	1584	2104	3688
Spec_cue	351	464	815
Speculated	509	611	1120
Family_history_of	0	16	16
Future	10	391	401
History_of	60	2736	2796
Hypothetical	9	113	122
Family_member	30	64	94
Other	78	388	466
Patient	3007	3950	6957
Total entities	37817	49220	87037
Nested entities	6153	8365	14518 (16.68%)
Discontinuous entities	347	210	557 (0.64%)
Total attributes	5606	10991	16597

Table 5. Distribution of annotated entities and attributes per type.

	Abstracts	EudraCT	Total
After	758	454	1212
Before	3118	2960	6078
Causes	1638	2181	3819
Combined_with	756	673	1429
Experiences	7783	9002	16785
Has_Age	254	573	827
Has_Dose_or_Strength	923	294	1217
Has_Drug_Form	186	84	270
Has_Duration_or_Interval	1149	1348	2497
Has_Frequency	370	167	537
Has_Quantifier_or_Qualifier	1935	3720	5655
Has_Result_or_Value	485	1640	2125
Has_Route_or_Mode	663	525	1188
Location_of	2833	4410	7243
Negation	1555	2050	3605
Overlap	3378	4499	7877
Speculation	488	601	1089
Used_for	2251	2513	4764
Total	30523	35181	68217

Table 6. Distribution of annotated relations per type.

	Abstracts	EudraCT	Total
Normalized entities	30459 (80.54%)	40714 (82.72%)	71156 (81.75%)
Not-normalized entities	7358 (19.46%)	8506 (17.28%)	15881 (18.25%)
Unclear normalization	1235 (3.27%)	1148 (2.33%)	2383 (2.74%)
Complex normalization	4223 (11.17%)	6158 (12.51%)	10381 (11.93%)
Total entities	37817	49220	87037

Table 7. Count of normalized entities.

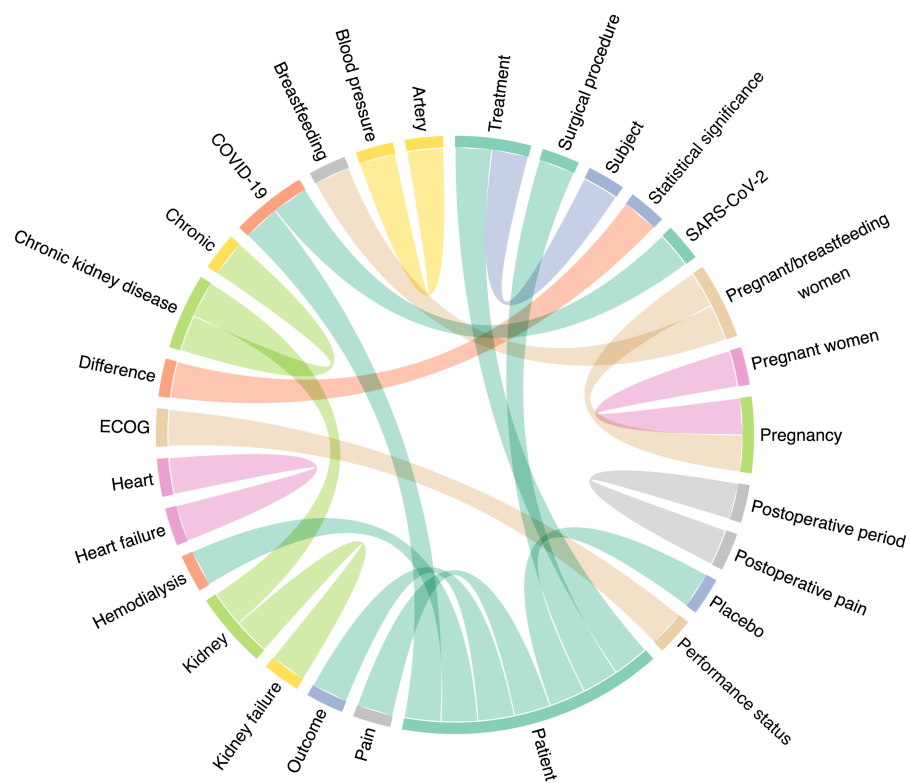


Figure 1. Chord diagram of the 20 most frequent relations between entities.

Relation Extraction Results

	Precision	Recall	F1	Accuracy
bert-base-multilingual-cased	0.884	0.874	0.879	0.917
	(±0.006)	(±0.003)	(±0.005)	(±0.001)
bert-base-multilingual-cased vs 2	0.880	0.876	0.878	0.916
	(±0.006)	(±0.005)	(±0.003)	(±0.001)
dccuchile-bert-base-spanish-wwm-uncased	0.868	0.857	0.862	0.907
	(±0.009)	(±0.006)	(±0.006)	(±0.003)
dccuchile-bert-base-spanish-wwm-uncased vs 2	0.877	0.854	0.864	0.908
	(±0.008)	(±0.005)	(±0.006)	(±0.003)
microsoft-mdeberta-v3-base	0.873	0.859	0.864	0.909
	(±0.010)	(±0.005)	(±0.005)	(±0.005)
microsoft-mdeberta-v3-base vs 2	0.886	0.857	0.869	0.911
	(±0.003)	(±0.007)	(±0.005)	(±0.003)
RoBERTa-bsc-bio-ehr-es	0.867	0.852	0.858	0.907
	(±0.007)	(±0.006)	(±0.006)	(±0.005)
RoBERTa-bsc-bio-ehr-es vs 2	0.875	0.858	0.865	0.910
	(±0.009)	(±0.004)	(±0.006)	(±0.004)

Table 2. Results of the relation extraction task (average $\pm$ standard deviation of 5 experimental rounds).

One-way analysis of variance					
P value	< 0.0001				
P value summary	***				
Are means signif. different? (P < 0.05)	Yes				
Number of groups	8				
F	8.594				
R square	0.6528				
Bartlett's test for equal variances					
Bartlett's statistic (corrected)	1.512				
P value	0.9819				
P value summary	ns				
Do the variances differ signif. (P < 0.05)	No				
ANOVA Table	SS	df	MS		
Treatment (between columns)	0.001849	7	0.0002641		
Residual (within columns)	0.0009833	32	3.073e-005		
Total	0.002832	39			
Bonferroni's Multiple Comparison Test	Mean Diff.	t	Significant? P < 0.05?	Summary	95% CI of diff
mBERT vs mBERT v2	0.001833	0.5229	No	ns	-0.01011 to 0.01378
mBERT vs BSC-BIO	0.01717	4.897	Yes	***	0.005220 to 0.02911
mBERT vs BSC-BIO v2	0.01333	3.803	Yes	*	0.001386 to 0.02528
mBERT vs BETO	0.02083	5.942	Yes	***	0.008886 to 0.03278
mBERT vs BETO v2	0.01417	4.041	Yes	**	0.002220 to 0.02611
mBERT vs mDeBERTa	0.01450	4.136	Yes	**	0.002553 to 0.02645
mBERT vs mDeBERTa v2	0.009167	2.615	No	ns	-0.002780 to 0.02111
mBERT v2 vs BSC-BIO	0.01533	4.374	Yes	**	0.003386 to 0.02728
mBERT v2 vs BSC-BIO v2	0.0115	3.280	No	ns	-0.0004470 to 0.02345
mBERT v2 vs BETO	0.01900	5.419	Yes	***	0.007053 to 0.03095
mBERT v2 vs BETO v2	0.01233	3.518	Yes	*	0.0003863 to 0.02428
mBERT v2 vs mDeBERTa	0.01267	3.613	Yes	*	0.0007195 to 0.02461
mBERT v2 vs mDeBERTa v2	0.007333	2.092	No	ns	-0.004614 to 0.01928
BSC-BIO vs BSC-BIO v2	-0.003833	1.093	No	ns	-0.01578 to 0.008114
BSC-BIO vs BETO	0.003667	1.046	No	ns	-0.008280 to 0.01561
BSC-BIO vs BETO v2	-0.003000	0.8557	No	ns	-0.01495 to 0.008947
BSC-BIO vs mDeBERTa	-0.002667	0.7606	No	ns	-0.01461 to 0.009280
BSC-BIO vs mDeBERTa v2	-0.008000	2.282	No	ns	-0.01995 to 0.003947
BSC-BIO v2 vs BETO	0.007500	2.139	No	ns	-0.004447 to 0.01945
BSC-BIO v2 vs BETO v2	0.0008334	0.2377	No	ns	-0.01111 to 0.01278
BSC-BIO v2 vs mDeBERTa	0.001167	0.3327	No	ns	-0.01078 to 0.01311
BSC-BIO v2 vs mDeBERTa v2	-0.004167	1.188	No	ns	-0.01611 to 0.007780
BETO vs BETO v2	-0.006667	1.902	No	ns	-0.01861 to 0.005280
BETO vs mDeBERTa	-0.006333	1.806	No	ns	-0.01828 to 0.005614
BETO vs mDeBERTa v2	-0.01167	3.328	No	ns	-0.02361 to 0.0002804
BETO v2 vs mDeBERTa	0.0003332	0.09504	No	ns	-0.01161 to 0.01228
BETO v2 vs mDeBERTa v2	-0.005000	1.426	No	ns	-0.01695 to 0.006947
mDeBERTa vs mDeBERTa v2	-0.005333	1.521	No	ns	-0.01728 to 0.006614

Table 7. One-way analysis of variance and Bonferroni's multiple comparison test between models for the F1 measure of the relation extraction task.

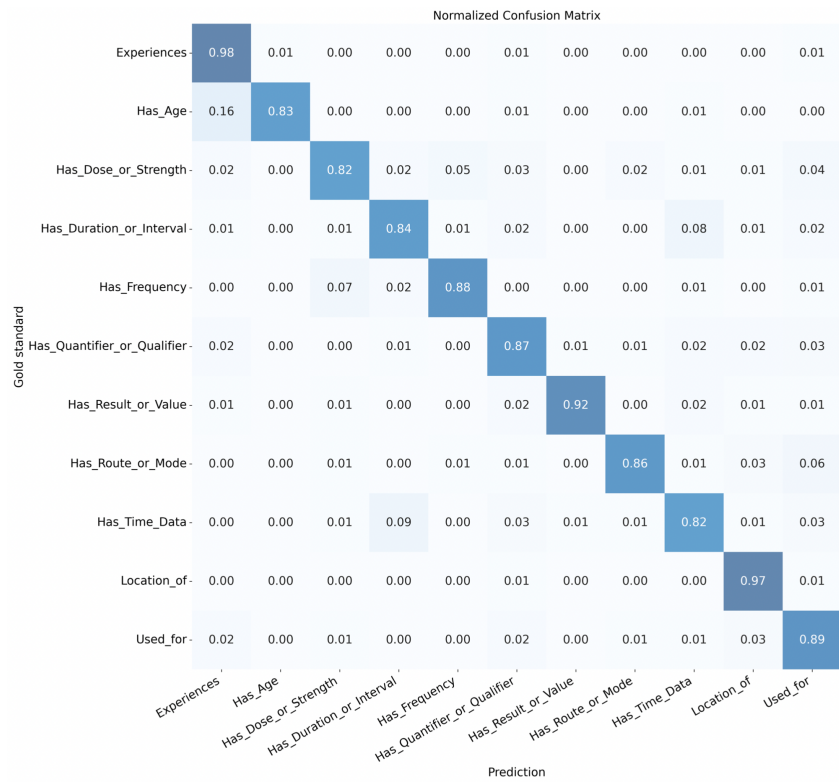


Figure 2. Confusion matrix of the bert-base-multilingual-cased vs 2 model.

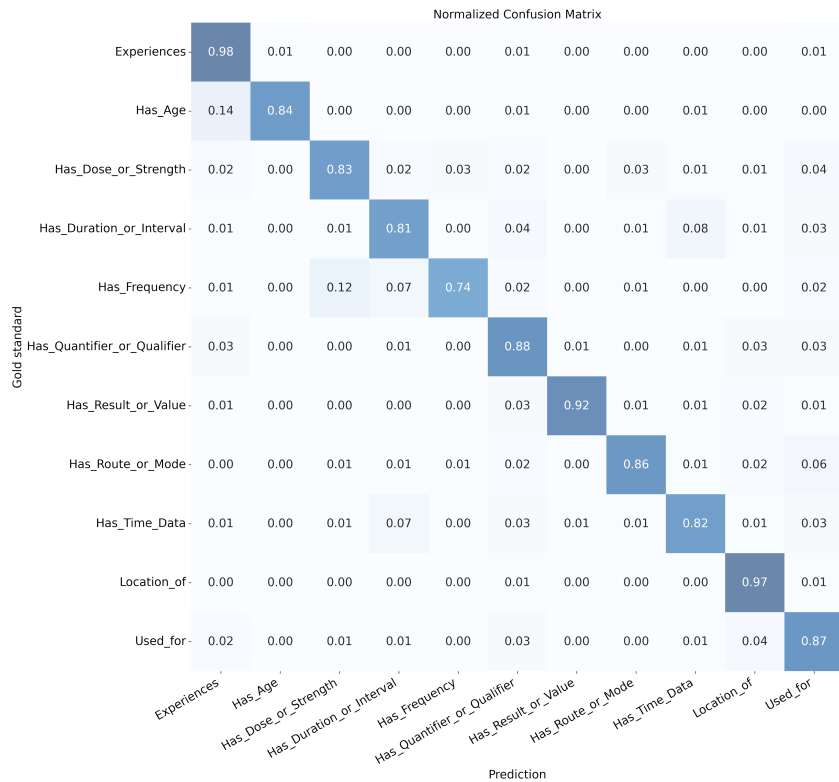


Figure 3. Confusion matrix of the dccuchile-bert-base-spanish-wwm-uncased vs 2 model.

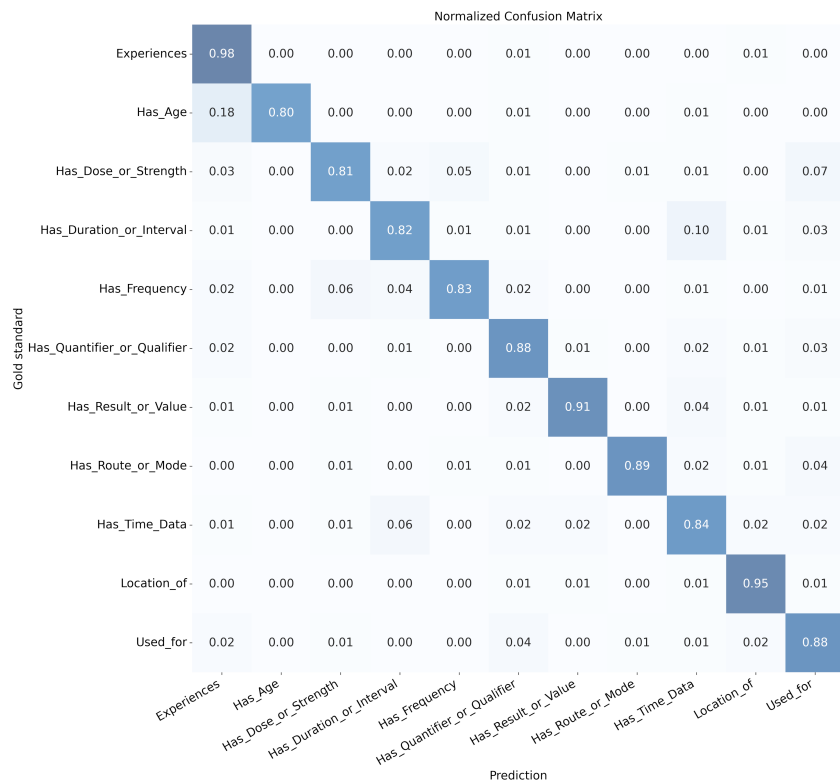


Figure 4. Confusion matrix of the microsoft-mdeberta-v3-base vs 2 model.

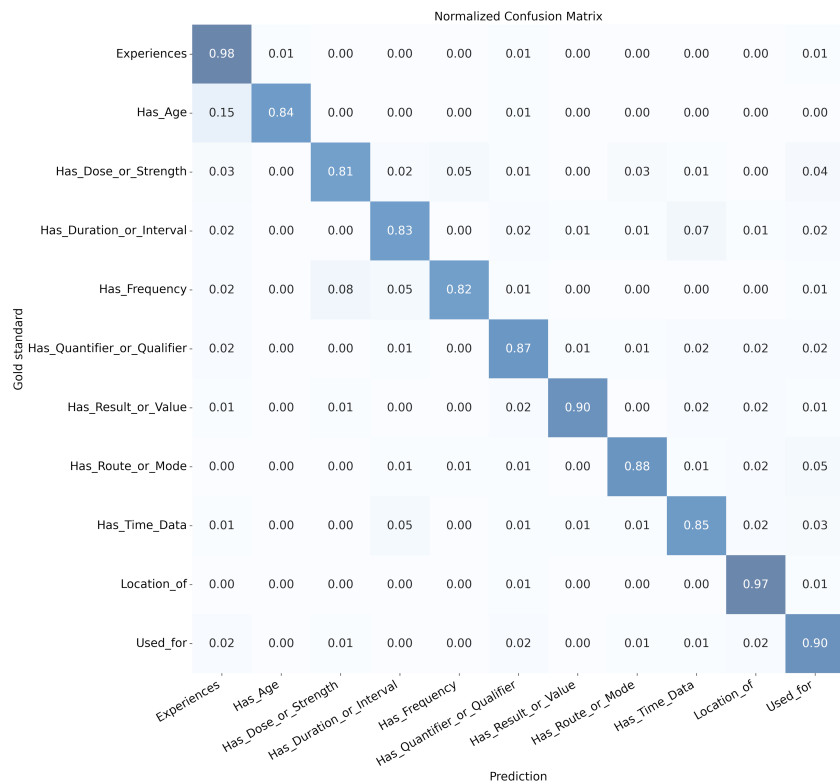


Figure 5. Confusion matrix of the RoBERTa-bsc-bio-ehr-es vs 2 model.

Explainability

We applied an explainable artificial intelligence (XAI) method to visualize the `bert-base-multilingual-cased` model's attention weights for specific input tokens (Figures S4-S7, Supplementary material). This analysis tends to show that Transformer models focus on key words in the medical entities from the sentence context to predict a relationship.

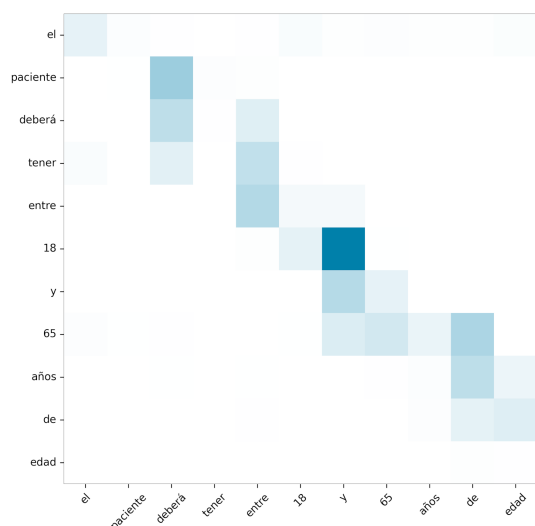


Figure 6. Attention weights predicting relation Has_Age in sentence 'The patient must be between the ages of 18 and 65'.

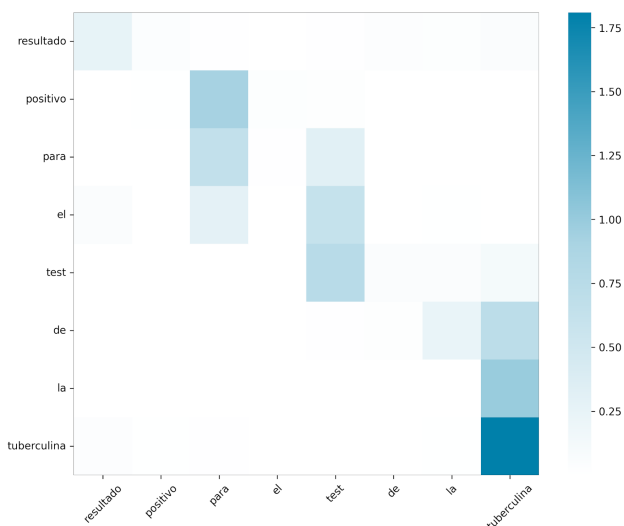


Figure 7. Attention weights predicting relation Has_Result_or_Value in sentence 'Positive result in the tuberculin test'.

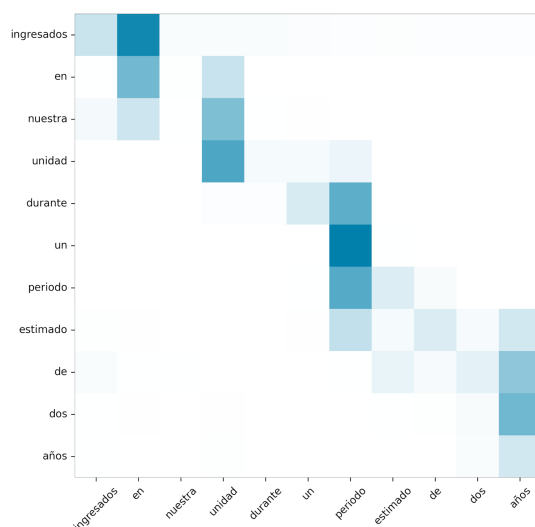


Figure 8. Attention weights predicting relation Has_Duration in sentence 'Admitted to our unit for an estimated period of two years'.

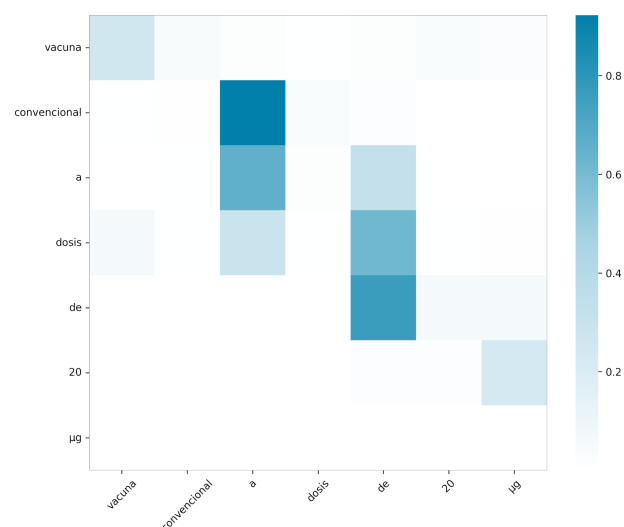


Figure 9. Attention weights predicting relation Has_Dose_or_Strength in sentence 'Conventional vaccine at a dose of 20 µg'.

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